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The Editor-in-Chief

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Dear Editor,

I submit the manuscript “MyUncleVision: entropy-guided, label-free discovery of unusual tissue organization in whole-slide images, with a maintenance-lattice order–entropy framework” for consideration as a research article in Computer Methods and Programs in Biomedicine.

Unsupervised discovery in whole-slide images needs to keep artifacts and routine histology out of its findings and to hand experts something they can check. The manuscript presents MyUncleVision, an open pipeline that places patch embeddings on a tissue lattice and measures order, entropy, state transitions and a three-axis novelty score, with definitions that generalize those of the maintenance-lattice framework MyUncle from two to d dimensions and reproduce them exactly at d = 2 (tested against the unmodified core). It ranks regions across slides, maps them, explains them with measured numbers, and runs from one command in a container.

The study is explicit about what it is. It is validated on synthetic slides with planted anomalies and artifacts, which are not tissue; the novelty score reaches an AUROC of 0.90 on them (slide-bootstrap interval 0.86–0.94) but the combination is not better than its best single axis, and the thresholds were adjusted after seeing synthetic results, so these are development results. The platform has not been run on kidney or any other real tissue beyond a file-format test, no pathologist has reviewed any output, and the manuscript specifies, without reporting, the evaluation protocol for the Kidney Precision Medicine Project images. I offer it as a methods and software paper whose real-tissue evaluation is the next step, and I would be glad to know if the journal prefers that evaluation to be part of the paper.

With the DINOv2 encoder the same synthetic benchmark gives an AUROC of 0.22 for the novelty score, below chance: at the default rarity scale of 1% the planted states (about 3.1% of the patches) are not rare, and DINOv2 embeds them more homogeneously than normal tissue, so two of the three axes point the wrong way; the appearance axis alone recovers when the scale is raised (AUROC 0.91 at 10%). With

DINOv2, 0 region(s) on slides with nothing planted entered a finding. The default settings are therefore not validated for DINOv2 on this benchmark, and a real run should compare encoders and inspect the axes separately.

The code, tests and results are open at https://github.com/sandlerleon/MyUncleVision. The manuscript is original, has not been published elsewhere and is not under consideration at any other journal. I am the sole author. A generative AI tool (Claude, Anthropic) assisted with writing and testing the code, running the analyses and drafting the text and figures; I reviewed and edited all of it and take full responsibility, as declared in the manuscript. I have no competing interests to declare and the work received no funding.

Thank you for considering this submission.

Sincerely,

Leon Sandler